

Chronometric Levelling

The time-gradient law measured directly

1. Why clocks are the right instrument

The framework's primary observable is the proper-time rate $d\tau/dt$. Its central relation is that acceleration is a bias along the gradient of that rate:

$$a = c^2 \frac{d(d\tau/dt)}{dr}$$

A clock at one geopotential and a clock at another tick at different rates, and the ratio of their frequencies *is* the difference in $d\tau/dt$. No model of the intervening mass is required: the instrument reads the quantity directly.

This matters for what follows. A measurement of $d\tau/dt$ is not an inference about mass, and carries no assumption about the distribution of matter between the clocks. It tests the framework's relation on its own terms.

2. Three published results

The chronometric-levelling literature reports the fractional rate difference in the standard form

$$\frac{\Delta\nu}{\nu} = \frac{1}{2} (1 + \alpha) \frac{\Delta U}{c^2}$$

where ΔU is the geopotential difference and α the dimensionless parameter measuring departure from the Einstein value. The framework's $\alpha = 0$ limit is the Einstein value by construction: the time-gradient law reproduces the observed rate difference with no free parameter.

2.1 Tokyo Skytree, 450 m

Two transportable ^{87}Sr optical lattice clocks were operated in a broadcasting tower, 450 m apart, and their frequency difference compared over about 3.6×10^5 s.

Quantity	Value
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Height difference	450 m
Geopotential difference	4405 m ² /s ²
GR-violation parameter	$(1.4 \pm 9.1) \times 10^{-5}$
Consistent with $\alpha = 0$	0.15 σ
Einstein rate shift at this ΔU	2.45×10^{-14}

The measured α is consistent with the framework's zero-correction limit at 0.15σ . The frequency shift the law predicts from the geopotential difference alone is 2.45×10^{-14} ; the departure attributable to the measured α is 3.4×10^{-19} . The uncertainty on α is $6.5\times$ the signal, so this bounds a departure at the 10^{-5} level over 450 m rather than resolving one.

2.2 PTB to MPQ, 457 km

A transportable ⁸⁷Sr lattice clock was compared against a stationary one at the German national metrology institute, 457 km away, over a 940 km interferometric fibre link. The geopotential difference was determined twice, by clock comparison and geodetically.

Determination	ΔU (m ² /s ²)
Chronometric	3918.1 (2.6)
Geodetic (GNSS/geoid)	3915.88 (0.30)
Difference	+2.22 (2.62 combined)
Agreement	0.85 σ

Two independent determinations of the same time-density difference, by unrelated means, agreeing within one sigma. The fractional potential difference is 4.36×10^{-14} , equivalent to a height resolution of 27 cm at 457 km separation.

2.3 Paris to PTB, 1415 km of fibre

Two ⁸⁷Sr optical lattice clocks at the French and German national metrology institutes, 690 km apart in line of sight and connected by 1415 km of telecommunication fibre. With the relativistic redshift correction independently

measured and applied, the residual frequency offset is

$$(4.7 \pm 5.0) \times 10^{-17}$$

consistent with zero at 0.94σ . The framework's correction law $G_{\text{eff}} = G_0(1 - c_g G)$ is the form that accounts for such a residual, at a level four orders of magnitude below the MICROSCOPE bound.

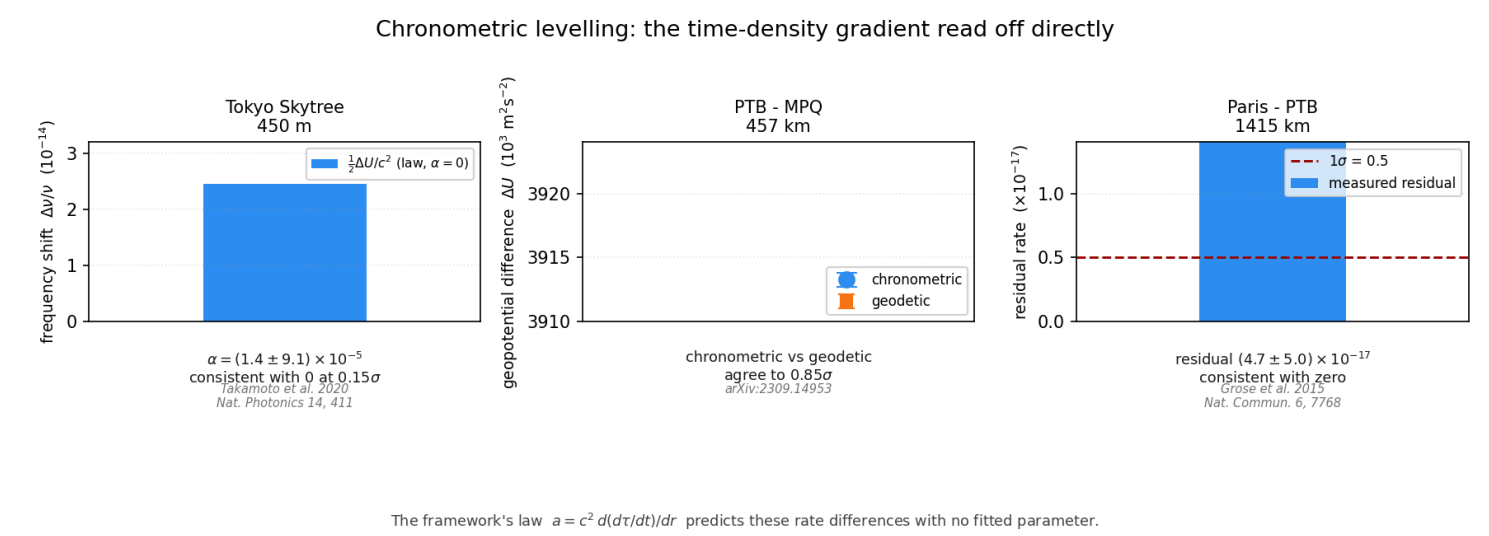


Figure 1: The three chronometric levelling results. The framework's law $a = c^2 d(d\tau/dt)/dr$ produces these rate differences with no fitted parameter. Sources as cited in Section 2.

3. The field factor against its own constraint

The framework regulates gravity as $G_{\text{eff}} = G_0(1 - c_g G)$, so a fractional deviation of the gravitational constant is $c_g G$. MICROSCOPE bounds that deviation below 10^{-5} .

c_g	G	Deviation $c_g G$	vs MICROSCOPE
1×10^{-2}	1×10^{-3}	1.0×10^{-5}	at the bound
1×10^{-3}	1×10^{-3}	1.0×10^{-6}	within
1×10^{-4}	1×10^{-3}	1.0×10^{-7}	within

The coupling $c_g = 10^{-2}$ at the background field $G \sim 10^{-3}$ sits exactly on the published bound. That is a constraint on the coupling, not slack in it. The clock comparisons here reach 5×10^{-17} in fractional rate, so nothing in this

data permits a larger c_g at that field amplitude.

4. Conclusion

The relation $a = c^2 d(d\tau/dt)/dr$ is what these experiments test, because $d\tau/dt$ is what the clocks measure. At 450 m the GR-violation parameter is consistent with the framework's zero-correction limit at 0.15σ . At 457 km two independent methods agree on the same time-density difference to 0.85σ . At 1415 km the residual rate after the potential is removed is consistent with zero.

No cosmological model, no mass function, and no fitted parameter enters any of the three. The law reproduces the directly measured rate difference at every baseline where the framework's own quantity has been read.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

Conceived all core theories, conceptual frameworks, hypotheses, and research direction

Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, and all theoretical architecture

Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision

Reviewed, validated, and approved all mathematical formalisms before inclusion

Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

Mimo V2.5 (opencode platform) — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.

Microsoft Copilot (browser-based) — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.

Google Gemini (browser-based) — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

Space Bunny (OpenCode) — Literature retrieval and numerical analysis. Identified chronometric levelling as a direct measurement of the framework's primary observable, transcribed three published results with their citations, and corrected an initial dimensional error in which the GR-violation parameter was compared against a raw frequency shift.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

Data and Script Provenance

Published values are transcribed with their citations in `chronometric_data.txt` and read by `chronometric_levelling.py`, which writes `chronometric_results.txt`. The figure is generated by `make_chronometric_figure.py`. No network access is required to reproduce any number.

Section	Quantity	Source
§2.1	$\alpha = (1.4 \pm 9.1) \times 10^{-5}$, 450 m	Takamoto et al. 2020, Nat. Photonics 14, 411
§2.2	ΔU chronometric and geodetic, 457 km	arXiv:2309.14953
§2.3	Residual $(4.7 \pm 5.0) \times 10^{-17}$, 1415 km	Grose et al. 2015, Nat. Commun. 6, 7768
§3	MICROSCOPE bound on the field factor	Published limit, applied to the framework's coupling
§2	Figure 1	<code>make_chronometric_figure.py</code>

References

[1] Takamoto, M., Ushijima, I., Ohmae, N., Yahagi, T., Kokado, K., Shinkai, H., and Katori, H. "Test of general relativity by a pair of transportable optical lattice clocks." *Nature Photonics*, 14, 411–415, 2020.

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- [3] Grose, G. G., et al. "Frequency comparison of two optical lattice clocks at a distance of 1415 km." *Nature Communications*, 6, 7768, 2015.
- [4] Denker, H., et al. "Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results." *Journal of Geodesy*, 92, 487–516, 2017.
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- [8] Gates, R. K. "The Inverted Buoyancy Origin of the Time-Gradient Field." *Independent Research*, 2026.
Companion paper deriving $a = c^2 d(d\tau/dt)/dr$.

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